EXAMPLE 9 Evaluate $\int e^{2x} \sec e^{2x}\, dx$.

SOLUTION We let

$$u = e^{2x}, \qquad du = 2e^{2x}\, dx$$

and proceed as follows:

$$\begin{aligned}\int e^{2x} \sec e^{2x}\, dx &= \tfrac{1}{2}\int (\sec e^{2x}) 2e^{2x}\, dx \\ &= \tfrac{1}{2}\int \sec u\, du \\ &= \tfrac{1}{2}\ln|\sec u + \tan u| + C \\ &= \tfrac{1}{2}\ln|\sec e^{2x} + \tan e^{2x}| + C\end{aligned}$$

EXAMPLE 10 Evaluate $\int (\csc x - 1)^2\, dx$.

SOLUTION

$$\begin{aligned}\int (\csc x - 1)^2\, dx &= \int (\csc^2 x - 2\csc x + 1)\, dx \\ &= \int \csc^2 x\, dx - 2\int \csc x\, dx + \int dx \\ &= -\cot x - 2\ln|\csc x - \cot x| + x + C.\end{aligned}$$

We shall discuss additional methods for integrating trigonometric expressions in Chapter 9.

EXERCISES 7.4

Exer. 1–36: Evaluate the integral.

1 (a) $\int \frac{1}{2x+7}\, dx$ (b) $\int_{-2}^{1} \frac{1}{2x+7}\, dx$

2 (a) $\int \frac{1}{4-5x}\, dx$ (b) $\int_{-1}^{0} \frac{1}{4-5x}\, dx$

3 (a) $\int \frac{4x}{x^2-9}\, dx$ (b) $\int_{1}^{2} \frac{4x}{x^2-9}\, dx$

4 (a) $\int \frac{3x}{x^2+4}\, dx$ (b) $\int_{1}^{2} \frac{3x}{x^2+4}\, dx$

5 (a) $\int e^{-4x}\, dx$ (b) $\int_{1}^{3} e^{-4x}\, dx$

6 (a) $\int x^2 e^{3x^3}\, dx$ (b) $\int_{1}^{2} x^2 e^{3x^3}\, dx$

7 (a) $\int \tan 2x\, dx$ (b) $\int_{0}^{\pi/8} \tan 2x\, dx$

8 (a) $\int \cot \tfrac{1}{3}x\, dx$ (b) $\int_{3\pi/2}^{9\pi/4} \cot \tfrac{1}{3}x\, dx$

9 (a) $\int \csc \tfrac{1}{2}x\, dx$ (b) $\int_{\pi}^{5\pi/3} \csc \tfrac{1}{2}x\, dx$

10 (a) $\int \sec 3x\, dx$ (b) $\int_{0}^{\pi/12} \sec 3x\, dx$

11 $\int \frac{x-2}{x^2-4x+9}\, dx$

12 $\int \frac{x^3}{x^4-5}\, dx$

13 $\int \frac{(x+2)^2}{x}\, dx$

14 $\int \frac{(2+\ln x)^{10}}{x}\, dx$

15 $\int \frac{\ln x}{x}\, dx$

16 $\int \frac{1}{x(\ln x)^2}\, dx$

17 $\int (x + e^{5x})\, dx$

18 $\int \frac{e^{\sqrt{x}}}{\sqrt{x}}\, dx$

19 $\int \frac{3 \sin x}{1 + 2 \cos x}\, dx$

20 $\int \frac{\sec^2 x}{1 + \tan x}\, dx$

21 $\int \frac{(e^x + 1)^2}{e^x}\, dx$

22 $\int \frac{e^x}{(e^x + 1)^2}\, dx$

23 $\int \frac{e^x - e^{-x}}{e^x + e^{-x}}\, dx$

24 $\int \frac{e^x}{e^x + 1}\, dx$

25 $\int \frac{\cot \sqrt[3]{x}}{\sqrt[3]{x^2}}\, dx$

26 $\int e^x(1 + \tan e^x)\, dx$

27 $\int \frac{1}{\cos 2x}\, dx$

28 $\int (x + \csc 8x)\, dx$

29 $\int \frac{\tan e^{-3x}}{e^{3x}}\, dx$

30 $\int e^{\cos x} \sin x\, dx$

31 $\int \frac{\cos^2 x}{\sin x}\, dx$

32 $\int \frac{\tan^2 2x}{\sec 2x}\, dx$

33 $\int \frac{\cos x \sin x}{\cos^2 x - 1}\, dx$

34 $\int (\tan 3x + \sec 3x)\, dx$

35 $\int (1 + \sec x)^2\, dx$

36 $\int \csc x\,(1 - \csc x)\, dx$

Exer. 37–38: Find the area of the region bounded by the graphs of the given equations.

37 $y = e^{2x}$, $y = 0$, $x = 0$, $x = \ln 3$

38 $y = 2 \tan x$, $y = 0$, $x = 0$, $x = \pi/4$

Exer. 39–40: Find the volume of the solid generated if the region bounded by the graphs of the equations is revolved about the indicated axis.

39 $y = e^{-x^2}$, $x = 0$, $x = 1$, $y = 0$; y-axis

40 $y = \sec x$, $x = -\pi/3$, $x = \pi/3$, $y = 0$; x-axis

Exer. 41–44: Solve the differential equation subject to the given conditions.

41 $y' = 4e^{2x} + 3e^{-2x}$; $y = 4$ if $x = 0$

42 $y' = 3e^{4x} - 8e^{-2x}$; $y = -2$ if $x = 0$

43 $y'' = 3e^{-x}$; $y = -1$ and $y' = 1$ if $x = 0$

44 $y'' = 6e^{2x}$; $y = -3$ and $y' = 2$ if $x = 0$

Exer. 45–46: A nonnegative function f defined on a closed interval $[a, b]$ is called a *probability density function* if $\int_a^b f(x)\, dx = 1$. Determine c so that the resulting function is a probability density function.

45 $f(x) = \frac{cx}{x^2 + 4}$ for $0 \le x \le 3$

46 $f(x) = cxe^{-x^2}$ for $0 \le x \le 10$

47 A culture of bacteria is growing at a rate of $3e^{0.2t}$ per hour, with t in hours and $0 \le t \le 20$.

(a) How many new bacteria will be in the culture after the first five hours?

(b) How many new bacteria are introduced in the sixth through the fourteenth hours?

(c) For approximately what value of t will the culture contain 150 new bacteria?

48 If a savings bond is purchased for \$500 with interest compounded continuously at 7% per year, then after t years the bond will be worth $500e^{0.07t}$ dollars.

(a) Approximately when will the bond be worth \$1000?

(b) Approximately when will the value of the bond be growing at a rate of \$50 per year?

49 The specific heat c of a metal such as silver is constant at temperatures T above 200 °K. If the temperature of the metal increases from T_1 to T_2, the area under the curve $y = c/T$ from T_1 to T_2 is called the *change in entropy* ΔS, a measurement of the increased molecular disorder of the system. Express ΔS in terms of T_1 and T_2.

50 The 1952 earthquake in Assam had a magnitude of 8.7 on the Richter scale—the largest ever recorded. (The October 1989 San Francisco earthquake had a magnitude of 7.1.) Seismologists have determined that if the largest earthquake in a given year has magnitude R, then the energy E (in joules) released by all earthquakes in that year can be estimated by using the formula

$$E = 9.13 \times 10^{12} \int_0^R e^{1.25x}\, dx.$$

Find E if $R = 8$.

51 In a circuit containing a 12-volt battery, a resistor, and a capacitor, the current $I(t)$ at time t is predicted to be $I(t) = 10e^{-4t}$ amperes. If $Q(t)$ is the charge (in coulombs) on the capacitor, then $I = dQ/dt$.

(a) If $Q(0) = 0$, find $Q(t)$.

(b) Find the charge on the capacitor after a long period of time.

52 A country that presently has coal reserves of 50 million tons used 6.5 million tons last year. Based on population projections, the rate of consumption R (in million tons/year) is expected to increase according to the formula $R = 6.5e^{0.02t}$, where t is the time in years. If the country uses only its own resources, estimate how many years the coal reserves will last.

53 A very small spherical particle (on the order of 5 microns in diameter) is projected into still air with an initial velocity of v_0 m/sec, but its velocity decreases because of drag forces. Its velocity after t seconds is given by $v(t) = v_0e^{-t/k}$ for some positive constant k.